

02456 Deep Learning

Exercises Week 1

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Prince 3.5 Prove that the following property holds for $\alpha \in \mathbb{R}^+$:

$$\text{ReLU}[\alpha \cdot z] = \alpha \cdot \text{ReLU}[z]$$

This is known as the *non-negative homogeneity property of the ReLU function*;

Essentially this is $\max(0, \alpha z) = \alpha \cdot \max(0, z)$. Now since only $z \in \mathbb{R}$ it is the only variable that can be negative (since $\alpha \in \mathbb{R}^+$). We can therefore say that if z is negative the expression is 0 and if z is positive the expression is αz , thereby $\max(0, \alpha z)$ which is $\text{ReLU}[\alpha \cdot z]$.

Prince 3.11 How many parameters does the model in figure 3.6 have?;

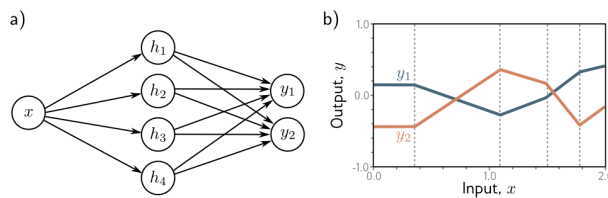


Figure 3.6 Network with one input, four hidden units, and two outputs. a) Visualization of network structure. b) This network produces two piecewise linear functions, $y_1[x]$ and $y_2[x]$. The four “joints” of these functions (at vertical dotted lines) are constrained to be in the same places since they share the same hidden units, but the slopes and overall height may differ.

This depends if we should count bias terms or not. They are not included in the visualization. Without bias terms we have the weights from x to the hidden layer = 4, and the weights from the hidden layer to the outputs y_1 and y_2 = 8. That is a total of 12 parameters. If bias terms are included we get 18.

Prince 4.1 Consider composing the two neural networks in figure 4.8. Draw a plot of the relationship between the input x and the output y' for $x \in [-1, 1]$;

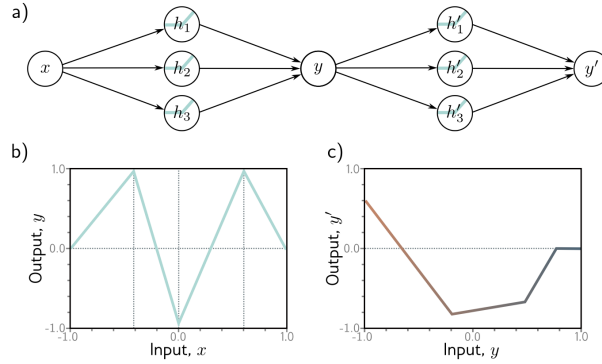
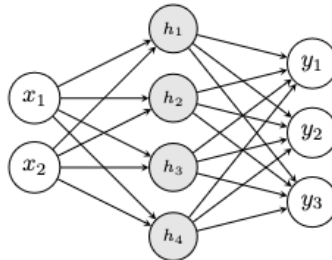


Figure 4.8 Composition of two networks for problem 4.1. a) The output y of the first network becomes the input to the second. b) The first network computes this function with output values $y \in [-1, 1]$. c) The second network computes this function on the input range $y \in [-1, 1]$.

Prince 4.11 Consider two neural networks that map a scalar input x to a scalar output y . The first network is shallow and has $D = 95$ hidden units. The second is deep and has $K = 10$ layers, each containing $D = 5$ hidden units. How many parameters does each network have? How many linear regions can each network make? Which would run faster?;

Assuming bias terms are included; the first neural network has $95x + 95y + (h + y) = 286$ since $x = y = 1$, $h = 95$ (i put the bias terms in parantheses to make a distinction). The second neural network has $5x + 9(5^2) + 5y + (h + y) = 286$ since $x = y = 1$, $h = 50$.

Exam 2025 Q1 Consider the fully connected neural network shown below, where $x \in \mathbb{R}^2$ is the input and $y \in \mathbb{R}^3$ is the output



Assume each neuron has a bias term. How many parameters does the model have in total;

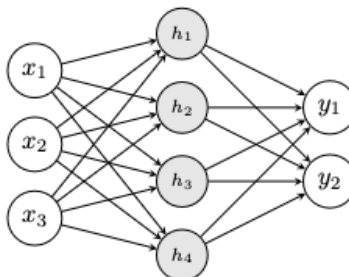
- A. 7
- B. 9
- C. 20
- D. 27

D. 27 is the correct answer. 8 weights from x to h , 12 weights from h to y and then 7 bias terms, $8 + 12 + 7 = 27$.

Exam 2025 Q2 Consider the fully connected neural network with one hidden layer defined as

$$f(x) = W^{(2)}\sigma(W^{(1)}x),$$

where $W^{(1)} \in \mathbb{R}^{n \times m}$, $W^{(2)} \in \mathbb{R}^{k \times l}$ and σ is the non-linear activation function. Assume that the network has the architecture shown below



What are the dimensions of the weight matrices;

- A. $n = 3, m = 4, k = 4, l = 2$
- B. $n = 4, m = 3, k = 2, l = 4$
- C. $n = 2, m = 4, k = 4, l = 3$
- D. $n = 4, m = 2, k = 3, l = 4$

B. is the correct answer. To map the 3×1 input vector to a 4×1 hidden layer vector we essentially perform $W^{(1)}x = h$. so for $n \times m \cdot 3 \times 1$ to be valid $n \times m = 4 \times 3$.